

The Challenge of Formative Assessment in Mathematics Education: Children's Minds, Teachers' Minds

Herbert P. Ginsburg

Teachers College Columbia University, New York, N.Y., USA

Key Words

Clinical interview · Developmental trajectories · Formative assessment · Learning difficulties · Mathematical competence

Abstract

The developmental psychology of mathematical thinking and the clinical interview method can make major contributions to education by transforming the process of formative assessment – the attempt to use information concerning student performance, knowledge, learning potential, and motivation to inform instruction. The clinical interview is a powerful method for learning about students' underlying cognitive competence. Cognitive developmental theory is essential because assessment is only as 'valid' as the ideas on which it is based. Yet current theories of developmental trajectories need to be supplemented by consideration of obstacles that the student must overcome. These involve the mathematics, instructional materials and the teacher.

Copyright © 2009 S. Karger AG, Basel

Researchers have made remarkable advances in understanding the development of children's mathematical knowledge [Clements & Sarama, 2007; Geary, 1994; Ginsburg, Cannon, Eisenband, & Pappas, 2006; Nunes & Bryant, 1996]. The question I explore in this paper is this: How can the new psychology of mathematical thinking be used to improve mathematics education? My answer is that one of its main contributions can be to inform, and indeed transform, the process of 'formative assessment,' which is usually defined [Heritage, Kim, & Vendlinski, 2008] as assessment designed to guide the teaching of students, particularly those who experience difficulty in learning.

I begin by defining the nature and goals of formative assessment, which focuses on performance, thinking/knowledge, learning potential, and affect/motivation, all

KARGER

Fax +41 61 306 12 34
E-Mail karger@karger.ch
www.karger.com

© 2009 S. Karger AG, Basel
0018–716X/09/0522–0109\$26.00/0

Accessible online at:
www.karger.com/hde

Herbert P. Ginsburg
Teachers College Columbia University
525 West 120th St., New York, NY 10027 (USA)
Tel. +1 212 678 3000, Fax +1 212 678 3837
E-Mail ginsburg@tc.columbia.edu

in the service of improving instruction. Next I show that formative assessment can employ three main methods for gathering data, namely observation, test, and clinical interview. I argue that the new research-based understanding of mathematical development can provide the conceptual framework for formative assessment, and that a clinical interview method modeled on Piaget's can be effective in providing essential data, especially concerning the usually unappreciated cognitive processes that underlie performance. Next I show how the resulting theoretically meaningful assessment can help to guide teaching. I conclude by proposing that a key challenge for mathematics education is to provide professional development designed to help teachers to understand and assess children's mathematical minds.

Formative Assessment: What Is It?

Formative assessment is generally defined as assessment for the purpose of instruction [Heritage et al., 2008]. The central idea is that assessment should not be reserved for an examination of achievement after the teacher has completed instruction. Rather, assessment should be used to gain information that can help the teacher plan effective instruction, particularly for the individual. Formative assessment can be *organized*, as when the teacher uses an already prepared observational instrument to learn about her students' motivation, or it can be *informal*, as when the teacher spontaneously questions students about their methods of solution.

What kind of information – gained by organized or informal methods – can be useful? The teacher needs to learn about performance, thinking/knowledge, learning potential and affect/motivation. Each of these corresponds to a major approach to psychology.

Performance refers to the child's overt behavior and achievement. The teacher simply wants to know whether the child is generally mastering the material, performing up to par, or 'getting it.' Is the child reasonably competent with the material being taught or totally lost? In most cases, the teacher is particularly interested in the latter possibility because it suggests the need for further intervention, for extra help. Several mathematics performance assessments involving 'curriculum based measurement' have emerged from the field of special education, driven as it is by a behaviorist orientation that stresses the need to help the struggling individual student [Fuchs, Fuchs, Karns, Hamlett, & Katzaroff, 1999; Gersten, Jordan, & Flojo, 2005].

Thinking/knowledge refers to the cognitive processes that underlie the child's overt performance. Wanting to know *why* the child is having trouble, the teacher typically looks for the answer in an analysis of weaknesses in the child's thinking: does he have trouble memorizing number facts or does he have important misconceptions about the mathematics being taught? I will argue below that the assessment should also include a focus on unsuspected strengths in the child's mathematical cognition. In any event, this approach, stemming from the cognitive traditions of Piaget and information processing, attempts to examine what is ordinarily not obvious, namely the child's mind.

Learning potential refers to the teacher's ultimate concern, namely whether the child is ready to learn the material in question. Some children whose performance is poor and who lack the relevant cognitive strategies or concepts have the potential to

learn them; other children do not. For Vygotsky, it is important to understand the child's current cognitive processes and knowledge, his 'actual developmental level' [Vygotsky, 1978, p. 85]. But it is even more useful for the teacher to gain insight into the level of '... potential development as determined through problem solving under adult guidance or in collaboration with more capable peers' (p. 86).

Finally the teacher must be interested in the child's *affect and motivation*, even if cognitive researchers typically pay only scant attention to such factors. It does not require research to know that at any age some children do not pay attention, are not interested, or fear mathematics, and that teaching requires dealing with these factors and other aspects of behavior management.

Each of these types of assessment – of performance, thinking/knowledge, learning potential, and affect/motivation – is based on psychological ideas and can be only as good as those ideas. If you are attempting to learn about the child's mathematical thinking and your only concept is 'intelligence,' you will learn little about back-up strategies or formal 'bugs,' and what you do learn – the child's level of 'intelligence' – will be of little concrete value in your efforts to teach the child. Fortunately, over the past 35 years or so, researchers in a variety of fields (particularly cognitive developmental psychology, educational psychology, and mathematics education) have conducted extensive investigations into young children's mathematical development, their ideas and skills in the areas of number, shape, space, pattern, and other mathematical topics. The literature is comprehensive and has been summarized in many extensive reviews [Baroody, Lai, & Mix, 2006; Clements & Sarama, 2007; Ginsburg et al., 2006; Nunes & Bryant, 1996].

My purpose here is not to attempt to summarize the work (I will present some examples below), but to claim that it offers many ideas of potential value for formative assessment. Teachers need assessments that give them a personally meaningful and practical 'theory' of the child's performance, thinking/knowledge, learning potential, and affect/motivation. Unless it is to be merely 'academic,' namely something learned in the academy and usually irrelevant for practice, the theory should make sense to the teacher, and should entail concepts that the teacher can see contribute to the practical job of teaching mathematics. It need not deal with the fine details that usefully interest cognitive researchers (like the myriad strategies that comprise young children's counting and addition), and it need not deal with broad generalities, like 'constructivism,' that may be of general theoretical relevance but offer little insight into the details of student behavior in an ordinary classroom. Instead, the teacher's theory should be mid-level, in the sense of dealing with concepts that lead to specific pedagogical maneuvers, and should avoid fine details and fancy talk.

Finally, the bottom line attribute of formative assessment is its *actionable* character: it is assessment that forms or informs instruction in a principled and effective manner. And if formative assessment can guide and improve teaching, what teacher would not be interested in it?

Methods of Formative Assessment

There are fundamentally three basic methods for assessment: observation, task, and clinical interview [Piaget, 1976].

Observation

This method has a long and distinguished history in psychology. In its purest form it involves the attempt to gather information about naturally unfolding behavior without influencing it. Thus, Charles Darwin recorded evidence of his child's emotional development [Darwin, 1955], and Piaget observed his infants' 'sensori-motor' development [Piaget, 1952b]. The intention was to learn how behavior unfolded 'naturally,' in the everyday context and especially in the absence of adult constraint or influence. The impulse is that of the naturalist, and the method of gathering data is active recording using some form of technology (for Piaget and Darwin paper, and now usually video).

Of course, observation is seldom as pure as the naturalist may prefer. It may be blended with other methods. Piaget did not only observe; he conducted miniature experiments, giving his infants various problems to solve. For example, if observations suggested that the child was capable of deliberate reaching for a toy, Piaget might alter its placement to see whether the child would accommodate her behavior to get it. Observation may take place in 'artificial' contexts like a laboratory or focus group. It can also include the gathering and subsequent analysis of *products*, like the child's written work in school or markings on the blackboard.

Much can be learned from keen observation. We may see that a child playing with blocks is exploring ideas of symmetry and pattern [Seo & Ginsburg, 2004], or that two children are having a conversation about what is the 'largest number' [Gelman, 1980]. Or we may see that the child's written homework shows in detail the steps taken to solve a problem.

At the same time, observational methods are not as simple as they appear. For one thing, observation is not passive – it is not just a matter of watching and learning from what one sees. Observation is a highly theoretical activity, depending on the availability of useful psychological ideas about mathematical thinking. As Piaget put it, '... if they are not on the look out for anything ... they will never find anything' [Piaget, 1976, p. 9]. Also observation is simply very hard to do in a classroom of 25 or 30 students. It's so hard that researchers use video recordings to assist in the process. But this is not a practical option for teachers.

But most importantly, observation is often mute with respect to student thinking. Spontaneous behavior may reveal little about the cognitive process underlying it. As Piaget put it, '... how many inexpressible thoughts must remain unknown so long as we restrict ourselves to observing the child without talking to him?' [Piaget, 1976, pp. 6–7].

Task or Test

Because behavior may not tell us all we want to know about the mind, we often employ another method of assessment, the task or test. This involves an adult-determined task that the child is asked to solve. The task is not a naturally occurring event; it is an adult contrivance, posing a problem from the adult perspective. The task may be the teacher's simple request that a child provide the correct name for a plane shape, or it might be an item on an achievement test. The teacher may use *standard administration* to give several children the same exact task. Or she may give different

children different tasks. In all these cases, the adult chooses the problem and the child has to deal with it.

Much can be learned from responses to tasks or test items. If all students receive the same items administered in a standard fashion, then we can learn about performance differences. Responses to the items may be of many types. Accuracy of response – the right and wrong answers – may be of interest, but so too may be the child’s verbal and nonverbal behavior, and even his strategies of solution. The teacher may ask the child to define a concept, or may observe what he writes on paper to solve a problem, thus obtaining information about strategy. Multiple-choice tests may be designed so that wrong answers clearly indicate well-known misconceptions or specific incorrect strategies of solution. Thus, the answer to this problem is not just wrong; it clearly identifies the bug of failing to ‘carry’ the appropriate tens value.

$$\begin{array}{r} 12 \\ +9 \\ \hline 111 \end{array}$$

At the same time, tasks and tests may not provide as much information as we would like and can use. Often a response to test or task is hard to interpret. Piaget [1952a] himself reported that when employed by the Binet laboratory to administer standardized intelligence tests, he soon found children’s responses to be of less interest than the cognitive processes that produced them but that could not be examined because of the method of standardized test administration. In everyday discourse, it is quite natural to ask the kind of follow-up question that is needed to illuminate responses to standard tests [diSessa, 2007]. When the child claims that $5 + 2 = 3$, what could be more appropriate than to ask, ‘Why do you think so?’ or ‘How did you figure that out?’ – both questions that are typically not appropriate for tests or tasks.

Clinical Interview

Because tasks and tests produce only limited responses, and because his interest centered around the investigation of cognitive processes, Piaget [1976] proposed use of the clinical interview, which, he felt, was designed to encompass the best features of task and observation. Typically, the interview begins with a task chosen by the adult. In this sense, the adult’s point of view dominates, at least at the outset. Then the interviewer observes the child’s response, behavior, affect and anything else that might be relevant, and develops an interpretation of the processes underlying the child’s behavior. To check the hypothesis, the interviewer develops new tasks as appropriate, follows up with questions designed to elicit thinking, and in general follows the child’s thought process to where it leads. In this sense, the interviewer makes every effort to be child-centered – to see the issues from the child’s point of view. The interviewer sometimes challenges the child’s response (‘But Johnny said that 2 and 2 is really 5.’) to assess its stability and the child’s confidence in it. The interviewer may sometimes employ gentle hints to help the child overcome a difficulty (‘Can you use your fingers to help you figure it out?’). The interviewer continues these nonstandardized interview maneuvers, which must be constructed on the spot, in real time,

until enough evidence has been obtained to support a reasonable interpretation of the child's behavior. In its purest form, the procedure deliberately does not employ standard administration; rather the procedure is customized for the individual child.

Like observation or task, the clinical interview is only as good as the theoretical ideas on which it is based. The interviewer must make interpretations of what the child says and does. Again, the interpretation is based on some kind of theory – explicit or intuitive – about children's thinking. The interview is not a mechanical process. It can succeed only if the interviewer produces sensible hypotheses and interpretations that can guide the search for evidence. And what makes the interview extremely difficult is that these interpretations must be produced quickly as the interviewer works with the child.

Here is an example of a very short and simple interview sequence with a first grader, Joe. Janet Eisenband was the interviewer, and this exchange occurred mid-way through the interview.

- Interviewer: How much is eight plus seven?
Joe: Eight plus seven equals ... [pause] fifteen.
Interviewer: OK. I have some questions for you. When you did eight plus seven, how did you do that?
Joe: Because eight plus eight equals sixteen and seven plus seven equals fourteen and if I add one more to that seven it would make, it would make fifteen. If I add one less on the eight, then it would make, then it would still make fifteen.

Despite its brevity, several features of the interview stand out. One is that Joe's initial pause suggested that his first response, fifteen, was not simply memorized but was the product of some thought. Second, a simple 'How did you do it?' type question elicited a very interesting result. The child's line of reasoning about $7 + 7$ can be considered an 'invented strategy,' a method that Joe might have developed himself to solve a simple 'number fact' problem. This kind of invented strategy typically employs something the child already knows (presumably he had memorized seven plus seven) along with a little adjustment (adding one) to solve the new problem. In Piaget's terms, using what one already knows to solve a new problem is assimilation, and adjusting one's prior knowledge to fit a new problem is accommodation. And the harmonious combination of the two is effective, 'equilibrated' thinking. Third, that his method was invented and not simply taught is suggested by the second half of his response where he claimed that he could get the same result if he were to 'add one less' to an eight in the double resulting in sixteen. This is remarkable because he appears intuitively to add a negative number rather than subtract a positive one. It would be highly unlikely that the negatives were taught at his grade level. And finally, it is of great interest to learn that Joe was able to express his methods of solution so clearly. Mathematics is in part a social enterprise and communication is at its heart [National Council of Teachers of Mathematics, 2000]. From a cognitive point of view, the development of metacognitive function is a key task for young children [Kuhn, 2000].

The fragment offered above basically involved only one basic question: 'When you did eight plus seven, how did you do that?' But interviews are seldom so simple. Accounts of more complex interviews are readily available [Ginsburg, 1997].

My claim is that the interview centering on the child's understanding of school mathematics is clearly an appropriate and powerful method for the teacher interested in formative assessment. I am not advocating that the teacher engage in interviews revolving around the specific problems, like conservation of numerical equivalence, in which Piaget had such great interest and which have revealed much of value. But interviews concerning issues of direct relevance for teaching should be at the heart of formative assessment.

If done well and guided by appropriate theory, clinical interviews can provide deep insight into thinking – typically deeper than that provided by observation, products, or tasks and tests – and can reveal strengths and weaknesses that otherwise may go undetected [Ginsburg, 1997]. The clinical interview can help the teacher surmount her own egocentrism and appreciate the child's: it can help her realize that children's thought may be different from her own, and that they may not in fact be learning what she is attempting to teach.

At the same time, the clinical interview is hard to implement. It depends on human skill, may not produce reliable results, and indeed done badly can distort student thinking and elicit only what the student thinks the interviewer wants to hear. But the existence of these difficulties does not necessarily imply that the method should not be used. (The very teaching of math is itself fraught with many difficulties and yet we do not abandon the effort. Indeed, interesting things are often hard to do and can be done badly.) Instead the difficulties require that care must be taken in training and supporting interviewers.

In brief, the clinical interview is a difficult but potentially rewarding method for investigating the child's mind [Ginsburg, 1997]. It is difficult because the interviewer must produce appropriate theoretical ideas about thinking and must possess the mental agility to develop useful questions on the spot. Yet the interview is rewarding because it can provide contact between the teacher's mind and the student's. For these reasons, the clinical interview method is one of Piaget's most important contributions to education.

Precursors of Formative Assessment: Mathematics and Three Trajectories

Formative assessment is built on a foundation of mathematical knowledge and understanding of three kinds of trajectories.

The Mathematics

Knowledge of mathematics – even 'profound understanding of fundamental mathematics' [Ma, 1999] – is essential for teaching and for assessment as well. The point is obvious: how can you teach something you do not understand? But what is not so obvious is that mathematical understanding is required to understand children's thinking. As Dewey [1976] put it many years ago: 'Really to interpret the child's present crude impulses in counting, measuring, and arranging things in rhythmic series, involves mathematical scholarship – a knowledge of the mathematical formulae and relations which have, in the history of the race, grown out of such crude beginnings' (p. 282).

Suppose that a child adds 23 and 46 mentally by first adding 20 and 40 to get 60; then adds 3 and 6 to get 9; and finally combines the numbers to get an answer of 69. The teacher who understands the mathematics in children's thinking [Ginsburg & Seo, 1999] – in this case, the place value ideas underlying the conventional algorithm – can appreciate that the child's mental addition makes use of key ideas about place value and base ten, and is in fact a reasonable approximation of the algorithm. Dewey's 'mathematical scholarship' is necessary to interpret the child's thinking in a way that is both insightful and practically relevant for teaching. Unfortunately, teachers in the USA have difficulty in understanding some basic mathematical concepts [Ma, 1999; Post, Harel, Behr, & Lesh, 1991; Zhou, Peverly, & Xin, 2006]. Their limited mathematical knowledge is likely to be as damaging for assessment as it is for teaching.

Developmental Trajectory

The next crucial aspect of formative assessment is teachers' understanding of three types of developmental trajectory. One involves *normative information* concerning performance. Suppose that a teacher interviews a first grader who cannot do any form of mental addition, even when the numbers are very small. Given her knowledge of what first graders can normally do, and given the available normative evidence, as contained for example in a norm-referenced test like the TEMA [Ginsburg & Baroody, 2003], the teacher is quite correct in concluding that the child is behind and needs extra help.

But norms are not enough: they need to be interpreted within a larger *cognitive trajectory*. Children begin by memorizing the first ten or so numbers (in English). They then struggle with the numbers from 13 to 19 (which, as we shall see, result from an irregular pattern). Next they begin to see patterns in the numbers above 20, and eventually realize that the 'decade numbers' (10, 20, etc.) are multiples of 10, and that the numbers from 1 to 9 can be added to each to form the correct counting system. Overall, the progression is from (a) rote memory to (b) detection of counting patterns, and then to (c) their use in generating numbers, and finally to (d) the understanding of counting patterns in terms of their underlying base ten composition and their place value representation. Of course, it is evident that trajectories describe a kind of average progression and do not necessarily capture the cognitive abilities of an individual child. Clements, Sarama and DiBiase [2004] propose that this cognitive progression needs to be considered in the context of specific teaching situations.

Third, equally important is the teacher's understanding of the *trajectory of mathematical ideas* in which the child's performance is embedded. For example, suppose that a first-grade child is struggling with the numbers above 30. He gets up to 39 without much difficulty, but cannot produce 40. Given that number, he can do 41–49 but not 50, and so on. The teacher needs to understand that the child's counting difficulty should not be treated as a minor, isolated event. Rather, it is part of a larger learning trajectory involving the development of base ten concepts. Counting to a 'high' number like 100 is important for first graders because it is one of the first instances in which they must deal with the systematic base ten property of numbers. Each spoken 'decades' number (20, 30 ...) is a multiple of 10, on to which are ap-

pended the units. Understanding these ideas underlies, or should underlie, the learning and teaching of counting (as well as reading and writing those same numbers). To appreciate the child's difficulty (and teach effectively), the teacher should understand that what appears to be a minor, isolated counting error may in fact have significance for the broader issue of base ten (and related place value) ideas.

In brief, formative assessment must rest on a foundation of mathematical understanding and can benefit from considering three kinds of developmental trajectory – performance, cognitive, and mathematical.

The Essential Elements of Formative Assessment: Obstacles and the Child's Mind

Understanding the mathematics and the developmental trajectories provide useful background for understanding the individual child. But next the teacher needs to consider three types of potential obstacles that stand between the child and the learning goal, and to appreciate the complexities of the child's mind.

Obstacles

Researchers have enjoyed great success in describing the development of the mathematical mind, but provide virtually no information on the obstacles it must overcome to function productively. Yet development and learning depend on overcoming barriers. In Piaget's view, the child assimilates and accommodates to various environmental obstacles that require further equilibration of existing cognitive structures for successful functioning [Piaget, 1952b]. A well-rounded and grounded psychology – especially a psychology of education – should offer an account of impediments to the student's learning of mathematics. I propose that some obstacles involve educational materials; some involve the teacher; and other obstacles involve the mathematics itself.

Textbooks, Manipulatives, and Other Teaching Materials. One of the biggest obstacles to learning mathematics is the textbook, which in the USA at least is often more colorful and lengthy than informative. Indeed, '... when mathematicians have reviewed recently published middle and high school textbooks, they have identified many errors and a large number of ambiguous and confusing statements and problems ...' [National Mathematics Advisory Panel, 2008, p. 55]. Textbooks are often repetitious from one grade to the next. '... [S]tudents in grades 2–5 encounter roughly 40–65% new content' [Flanders, 1987, p. 20]. Some repetition is no doubt valuable, at least for some students, but too much can be deadly boring. Further, compared with Chinese teachers, American teachers receive little help from teacher guides in how to implement a text [Li, Ding, Capraro, & Capraro, 2008]. When a child is not learning, the teacher might locate at least part of the difficulty in the large, brightly colored, but occasionally inaccurate text that the student is required to study. Sometimes the text is not so much inaccurate as it is misleading. For example, teachers attempt to teach the notion of equivalence in the first few grades and beyond. In this view, statements like $5 = 3 + 2$ and $5 = 5$ are valid expressions of a numerical equivalence.

lence relationship between the two sides of the equation. But textbooks very seldom provide examples of this type. Instead, the child is likely to see, over and over again, statements like $3 + 2 = 5$ [Seo & Ginsburg, 2003], which feed into the typical child's misconception that the '=' sign refers not to equivalence, but to an operation. In this view, the equals sign means, 'get the answer' or 'makes,' and it is very hard to teach the child otherwise [Baroody & Ginsburg, 1983]. Indeed, the problem persists into the middle school years [Knuth, Stephens, McNeil, & Alibali, 2006]. Manipulatives may present difficult challenges for children. They do not guarantee learning [Clements & McMillen, 1996], and some may work well in some mathematical contexts but not others. For example, a number line is very effective in showing the *operations* of addition or subtraction and their inverse. If you want to add 2 and 3, you start at 0, jump two spaces forward and then another three, and land on 5. If you want to subtract 3 from 5, you start at 5 and jump 'backwards' 3 spaces and there you are at 2. This is a very clear demonstration of the operations of addition and subtraction. But how do you demonstrate on the number line the equivalence between 5 and $2 + 3$? You could use two number lines, indicate a 5 on one, perform the operations described above for the other, and then observe that both number lines show the 5. But the effect is not as clear as could be achieved by using a balance with 5 objects on one side and a group of 2 and a group of 3 on the other. At the same time, the balance is not as effective in showing the addition and subtraction operations as is the number line.

The Teacher. Teachers may contribute to the problem as well. Despite their claim that they teach equivalence, many teachers, like their textbooks, present the '=' sign almost entirely in the context of solving computational problems and seldom refer to equivalence. An even more serious problem of course is that mathematics teaching, at least in US schools, is often of poor quality and fails to stress thinking. '... [E]xperiences in fifth grade, although highly variable, were geared toward performance of basic reading and math skills, not problem-solving or reasoning skills or other content areas. Few opportunities were provided to learn in small groups, to improve analytical skills, or to interact extensively with teachers. This pattern of instruction appears inconsistent with aims to add depth to students' understanding' [Pianta, Belsky, Houts, & Morrison, 2007, p. 1796]. To understand the obstacles to learning, teachers need to examine their own teaching practices. This kind of reflection can be facilitated by 'lesson study' [Fernandez, 2002] or other organized 'communities of learners' [Bransford, Brown, & Cocking, 1998], in which teachers get the opportunity to consult with peers in a common effort to understand and improve teaching.

The Mathematics. The mathematics itself or its representation can present obstacles to overcome. For example, in learning the numbers from 13 to 19, the main obstacle for children speaking many Western languages is that the spoken numbers are poorly designed. 'Eleven' and 'twelve' are effectively nonsense syllables for English-speaking children, and the numbers from 13 to 19 are said as a unit number followed by a ten, whereas numbers from 21 to 99 are said as they are written, namely a ten followed by a unit. Thus, we say 'thirteen' (or 'three-ten') which is the reverse of 'twenty-three' ('two-ten-three'). By contrast, the Chinese counting words and other East Asian counting systems based on the Chinese are elegantly designed [Miller & Parades, 1996]. After 'ten,' these languages say the equivalent of 'ten-one, ten-

two ...' and are perfectly consistent through 'nine-ten-nine.' By contrast, the English-speaking child (and children speaking other Western languages) first has to learn 12 counting words by rote memory (the Chinese child need learn only 10); then learns the backwards and irrational structure of the numbers from 13 to 19; and next has to unlearn that structure to acquire the finally sensible method of patterning the numbers from 21 to 99. Although for the English-speaking child the higher numbers are paradoxically easier to learn than the lower, teachers are sometimes reluctant to teach the larger numbers on the misguided grounds that they are too hard for young children. Another example involves various forms of mathematical symbolism. The '-' symbol refers to the operation of subtraction and to negative numbers. The '=' sign is used in situations in which an operation is meant to be performed as well as in situations where the issue of equivalence is of primary concern. Arithmetic operations are sometimes written vertically and sometimes horizontally. Vertically written statements do not involve the '=' sign; horizontally written statements do. What is the naïve child supposed to make of these complexities? And then of course some mathematical ideas can be difficult in themselves. For example, in one sense, absolute value, -10 is as 'big' as $+10$. But from a child's point of view, 10 is much 'bigger' because it is a positive number – you had to 'take away' a lot from $+10$ to get to -10 on the number line.

In brief, it is clear that obstacles of various sorts – including the teaching materials, the teacher, and the mathematics itself – can stand in the way of the child's learning mathematics in school. Yet we have a great deal to learn about the nature and operation of these obstacles.

The Child's Mind

Finally, having considered the mathematics to be taught, the performance, cognitive, and mathematical learning trajectories, and the obstacles posed by educational materials, teaching practices, and the mathematics itself, formative assessment examines what the individual child brings to the task. This is of course the area to which cognitive research can make enormous contributions. Researchers have produced comprehensive accounts of the mathematical development, from basic understanding of arithmetic [Baroody & Dowker, 2003] to rational number [Lamon, 2007]. The research examines the everyday mathematics that children bring to school [Levine, Jordan, & Huttenlocher, 1992] or develop even when they do not attend school [Nunes, Schliemann, & Carraher, 1993], and the mathematics that they develop in school, for example their invented strategies [Ginsburg, 1989], their 'bugs' [VanLehn, 1990], and their understanding and metacognition with respect to mathematics [Schoenfeld, 1987]. The research deals with individual differences [Dowker, 2005] as well as cognitive universals [Klein & Starkey, 1988].

Formative assessment can benefit a great deal from attending to and measuring these important aspects of mathematical cognition. Research on mathematical thinking makes important contributions to education when it provides the specific ideas that can enormously expand teachers' minds so that they can understand children's. The new insights can help teachers advance beyond the slogan that understanding children's thinking is important and to abandon vague and reverential stage notions ('According to Piaget, this child is preoperational and therefore can't

understand abstract math’’) in favor of specific analyses that show promise of guiding instruction.

At the same time, an important piece is missing from the cognitive psychology. The research seldom considers the relationships among various aspects of children’s mathematical thinking. Researchers tend to focus on isolated aspects of children’s knowledge. They study addition strategies or understanding of geometric forms or appreciation of the ‘=’ sign. Each of these is interesting, but does not provide a picture of the ‘cognitive whole child’ (let’s forget about the learner’s personality for the moment).

But Vygotsky’s theory provides a useful framework for thinking about the relations among various aspects of the child’s mathematical knowledge. He proposed that children enter school with various ‘spontaneous’ concepts, very much like the everyday mathematics already described. These spontaneous concepts are very different from ‘scientific’ or academic concepts – here mathematics – as taught in school. ‘The strength of scientific concepts lies in their conscious and deliberate character. Spontaneous concepts, on the contrary, are strong in what concerns the situational, empirical, and practical’ [Vygotsky, 1986, p. 194].

Vygotsky proposed that everyday knowledge may contain within it important ideas that can serve as the basis for understanding more formal and general notions. Hence one important educational strategy is to relate the spontaneous and scientific, the everyday and academic. For example, the child needs to see that his informal ideas about adding are related to the addition taught in school. The child already knows, without the benefit of schooling, that adding something to something else makes more. The child can then use this idea to understand what happens when one does written addition in school, and indeed can use his everyday idea to check the results of written computation.

Of course there may be occasions when a synthesis would not be particularly useful, particularly when the child’s everyday knowledge is faulty, as when he suffers from various misconceptions. But the general idea is important: for learning to be meaningful, the child must comfortably relate his personal knowledge to what is taught in school. He can benefit from organizing and systematizing what he already knows, and from linking the richness of his prior knowledge to the powerful abstractions taught in school. Many students do not do this. They feel that what they learn in school is merely ‘academic,’ in the pejorative sense of unrelated to anything personal and sensible, like everyday mathematics.

In the spirit of Vygotsky’s concept of synthesis, I propose that mathematical *profiles* may provide insight into important relationships among various aspects of mathematical knowledge. At present, the system is a conjecture. Yet it seems to capture something important about children’s mathematics learning in school; teachers feel that it makes sense – it has an intuitive appeal. We are conducting systematic research on the issue.

I propose four categories of mathematics learners. The categories are general, although the specific examples involve number facts and arithmetic.

The expert remembers her number facts and can efficiently apply procedures like multiplication with regrouping. This fluency and proficiency warms the hearts of those advocating a back to basics policy. At the same time, the expert understands the facts and procedures in two senses: she can provide the ‘official explanation’ as offered by the teacher or the text, for example, ‘Eight times seven gives the same re-

sult as seven times eight because multiplication is commutative.’ But this explanation is more than parroted words. She relates it to her own everyday knowledge: ‘If you have an egg carton with 12 eggs, you can turn it to any angle and it is still 12.’ Moreover, if memory fails, she uses backup strategies to produce the needed result. She thinks of mathematics as a meaningful activity requiring thought. And finally, she is aware of her thinking and can describe it to others. She has constructed mathematical knowledge that includes syntheses between various aspects of her personal knowledge and formal, mathematical concepts. True experts may be rare.

The mechanical learner, the master of illusory knowledge, remembers number facts and also employs procedures with efficiency. As a result he scores well on various standard tests and is considered to be a high achiever. But in fact, he understands little. He cannot make sense of much mathematics, and may indeed not believe that doing so would be a good thing. He may parrot the teacher’s explanation, but when pressed does not really understand it, either in purely formal mathematical terms, or with respect to his own everyday knowledge. What’s most important is doing well on the tests. The mechanical learner does the problems in the way he thinks the teacher wants. He tries to outfox the teacher by studying her expression to figure out what answer she expects [Holt, 1982]. This kind of learner is probably very common and may indeed be the ideal for many teachers – but not for those whose goal is sense-making and critical thinking.

The ordinary student performs reasonably well but makes mistakes. She remembers many facts and can apply procedures more or less accurately. She is perhaps occasionally sloppy and slothful. But she has a reasonable understanding of the basic formal ideas and can even explain some of them. She can accept that mathematics ought to make sense. She also has many everyday concepts and procedures, but does not always connect them with what is taught in school. Her underlying competence is sound, despite the fact that she often does not exhibit her knowledge to good effect and may perform at a mediocre level. In crude terms we can say that she is smarter than she looks. Perhaps there are many students of this type. They are ripe for good teaching.

The lost mathematical soul seems to have neither skill nor understanding. He does not remember facts or procedures, and he does not seem to understand what he is doing. Maybe he has various forms of everyday mathematical knowledge, but he does not exhibit them in school, and does not connect them with what he learns there. He does not see mathematics as a meaningful subject: it is rather a major ordeal, a minor torture. But we must be cautious about this diagnosis. We have to keep looking. Almost all children do have some competence, if only we knew how to help them find and express it. In other words, the problem may lie in our ability to assess the child’s knowledge, not in his lack of it.

These profiles seem to make sense to teachers and others, and to capture something fundamental about our experiences with school learning. In particular, they present teachers some new ways to think about mathematics learning. One is that expertise involves far more than ordinary achievement as measured by conventional tests. It requires integrating everyday and formal knowledge, and a sense-making approach to mathematics. A second is that the mechanical learner may do well but his success is illusory. Children who seem to know something may not. The teacher needs to deepen their understanding. A third is that children who seem to be getting by may know much more than meets the eye. These children may seem

not to know a great deal, but really do. And finally, there are indeed lost mathematical souls, but we must be cautious in identifying them: they, too, may know more than is initially apparent.

The Less than Royal Road to Teaching

Suppose then that the teacher conducts the kind of formative assessment described above. She examines the obstacles facing a child and attempts to learn about her everyday and academic knowledge and the relations between them. She sees that a child seems to fit, more or less, into a certain profile. What next? How does the formative assessment lead to improved teaching? There are several ways it can help and several ways it can't.

Defining the Teaching Goals

First, the cognitive analysis helps to frame the general and specific *goals* of mathematics education. On a general level, the ideas underlying the formative assessment suggest that the goal of mathematics education is not only to produce good performance, but also to help create in the child an understanding that results from synthesizing the everyday and the academic, the informal and formal. For many teachers, this is a radically different goal for mathematics education.

The assessment also uses information about obstacles and mathematical knowledge to specify more limited and actionable goals. On the one hand, the assessment identifies specific obstacles presented by specific learning tasks. The child learning to count must confront the partly irrational but fundamentally rule-governed English counting system. The child learning to do addition with regrouping must learn to decipher the written symbolism of the place value system. The child learning about equivalence must learn to overcome the limited use of the '=' sign in textbooks and the teacher's everyday practice. Obstacles vary across mathematical topics.

On the other hand, the assessment reveals that individual children, or children falling into various profiles, bring different skills, everyday and formal, to the task. This is a major insight for teachers who are often unaware of the everyday knowledge that can be exploited to promote school learning.

If obstacles differ among tasks, and if skills and knowledge differ among children, then different instructional goals are appropriate for different children. The formative assessment can help the teacher to specify these goals.

So the broad goal is good performance and also meaningful understanding, often based on the synthesis of the everyday and academic. The specific goals are overcoming the different obstacles inherent in different topics, and meeting the different needs of children with different profiles. To the extent that it can specify the obstacles and the child's everyday and academic knowledge with respect to a specific topic, a system of formative assessment can point teachers towards specific learning problems that need to be addressed. The inspiring slogans of mathematics education do not provide this kind of specificity, and neither do mathematics achievement measures.

The Difficulty of Teaching and Some Guiding Principles

How can the teacher help different types of students overcome the obstacles and create the meaningful synthesis of what is known and to be learned? First, I suggest a general perspective on pedagogical methods. Then I point to two broad pedagogical principles, one drawing heavily on Vygotsky and the other on Piaget.

There Is No Royal Road to Pedagogy. There is no single, foolproof way to teach early mathematics or almost anything else. Analysis of teaching young children mathematics [Ginsburg & Amit, 2008] reveals a wide variety of useful methods, ranging from discovery to drill, from humor to pressure, and from telling to modeling. The same is probably the case at all levels, from preschool through graduate school. Teaching is not easy, and in fact on any one occasion in any classroom it is extremely likely to fail: how can one teacher possibly meet all the needs of a relatively large number of students? Teaching is an extraordinarily difficult profession, in which the best one can hope for is a modicum of success. (The same is true in baseball, where excellence involves failure on two of three attempts.) In any event, achieving the specific goals indicated by the assessment requires many different specific methods, some perhaps a matter of personal style. But there are at least two general principles that can help.

Mathematize. Sometimes children's everyday knowledge or their invented strategies contain within them the seeds of ideas that are crucial for formal, academic knowledge. For example, we [Ginsburg & Seo, 1999] observed a second-grade teacher who asked her children to solve a problem they had not seen before, 23×4 , written vertically. First they sat at their desks, trying to work out the problem on paper. They used many different methods. Then the teacher selected several children (chosen because their methods differed, even though all got the right answer) to write their methods of solution on the blackboard. One girl (fig. 1) wrote a column of four 20s and added them to get 80. She then added the four 3s in the same way and got 12. Next she added the 80 and 12 to get the correct answer, 92. Her invented addition strategy for solving the new problem was effective, but the goal was to teach her multiplication. One possibility for doing this would be to talk about how she converted the multiplication problem into addition by using several kinds of knowledge. She decomposed 23 into tens and units; she then added the tens and units separately. The invented strategy provides the opportunity for explicit discussion – that is for *mathematizing* or casting her work into explicitly mathematical form – of composition and decomposition; the base ten system; the place value method for representing it and the relation between addition and multiplication. The work of another child can be used to move the discussion even further. A boy (fig. 2) broke up the problem into several multiplication components, 4×20 , 4×3 , and even 4×0 . He then added the products to get the correct answer. This invented strategy provides a bridge from that used by the first child. Instead of adding the four 20s, and the four 3s, as the girl did, the boy multiplied them. What a splendid opportunity to talk about the relations between addition and subtraction. But even more important is the boy's decomposition of the problem into separate multiplications of the tens and units, and then the addition of the resulting products. (The multiplication of zero is a bit of a mystery: maybe he did it because he had multiplied 4×20 and wanted to break that down

$$\begin{array}{r}
 23 \\
 + 23 \\
 \hline
 46
 \end{array}
 \qquad
 \begin{array}{r}
 23 \\
 + 23 \\
 \hline
 46
 \end{array}$$

$$\begin{array}{r}
 46 \\
 + 46 \\
 \hline
 92
 \end{array}$$

Fig. 1. Multiplying by adding.

$$\begin{array}{r}
 0 \\
 \times 4 \\
 \hline
 0
 \end{array}
 \qquad
 \begin{array}{r}
 20 \\
 \times 4 \\
 \hline
 80
 \end{array}$$

$$\begin{array}{r}
 3 \\
 \times 4 \\
 \hline
 12
 \end{array}
 \qquad
 \begin{array}{r}
 80 \\
 12 \\
 + 0 \\
 \hline
 92
 \end{array}$$

Fig. 2. Decomposing multiplication.

into tens and units as well.) In effect, the boy came very close to discovering the standard algorithm, and the teacher can use his work to introduce formal ideas in a meaningful way. That is, the teacher can mathematize his work, or in Vygotsky's terms synthesize what the child already knows (his invented method of multiplication) with the formal ideas of mathematics.

Create Cognitive Conflict. Sometimes children's assimilation of new material into what they already know results in 'bugs,' as when the child believes that

$$\begin{array}{r}
 12 \\
 -9 \\
 \hline
 17
 \end{array}$$

because you can't subtract a smaller number from a larger. The child has taken a principle that might be useful in certain contexts and generalized it in an inappropriate manner. In this case, the assessment reveals a misconception that needs to be corrected. Cognitive conflict, which lies at the heart of development [Piaget, 1985], can be useful in dealing with misconceptions [Limon, 2001]. For example, the teacher can ask the child described above to solve the problem by using blocks or some other countable material. Presumably by the time the child is asked to deal with subtraction problems involving regrouping he should be able to use the blocks to determine that taking away 9 from 12 leaves 3. (If he cannot use the blocks in this way, he shouldn't be learning subtraction with regrouping.) If the child succeeds, then the teacher can ask him to compare the results of the two operations, one on paper and the other with blocks. Most children tend to believe the evidence provided by ma-

nipulation of real objects like blocks, and therefore face a conflict. Which one is correct, the obviously true result in the real world, or the written result in the teacher's world, the strange new world of formal mathematics? The cognitive conflict needs to be resolved, and the teacher can take advantage of it to interpret the algorithm in terms of the successful block methods that the child uses. Indeed, manipulatives can play a special role in creating a conflict between the child's written mathematics and the concrete objects with which he works. Yet, for this approach to succeed, the child needs to learn that the manipulatives and the written work should yield the same results. That is, the child needs to be convinced written mathematics should make sense. Not all children believe this, especially as they gain experience with schooling. I have seen some children cheat with manipulatives in order to make them result in the same answer produced by written work.

The Teacher's Intermediary Inventive Mind

Of course, pedagogical principles other than mathematizing and introducing conflict can also serve a useful purpose. For example, drill and practice can be useful in certain situations (when a child understands but needs to memorize) and so can telling [Schwartz & Bransford, 1998] or lecturing (when a child has had ample concrete experience and can easily relate a spoken explanation to it).

But the main point is that the assessment does not by itself result in specific instructional activities. Instead, it identifies obstacles and mathematical thinking processes, and general pedagogical principles point the way towards broad approaches to teaching. The specific teaching methods used with a particular child must depend on the teacher's 'intermediary inventive mind.' As William James [1958] put it many years ago:

You make a great, a very great mistake, if you think that psychology, being the science of the mind's laws, is something from which you can deduce definite ... methods of instruction for immediate schoolroom use. Psychology is a science, and teaching is an art; and sciences never generate arts directly out of themselves. An intermediary inventive mind must make the application, by using its originality. (pp. 23–24)

Think of it this way: the teacher, understanding the mathematics and the three kinds of trajectories (performance, cognitive and mathematical), learns about the obstacles a child faces in learning a particular topic and about the intellectual resources that she brings to the task. With this knowledge in mind, and relying on some general principles of instruction, like mathematizing or creating intellectual conflict, the teacher then must use her own 'intermediary inventive mind' to decide upon a specific course of action with a particular child, or group of children. She may make use of some textbook or activity or game or manipulative. But these materials in themselves do not guarantee a solution to the problem. At best they provide tools with which the teacher can work. But she has to figure out how to use them effectively in a specific case. She needs to use her practitioner's knowledge, her inventive mind, and her stylistic preferences to 'make the application' of the general principles to the particular case. Any material can be used badly or well. Manipulatives may help or harm, and the same is true of textbooks. Good activities are plentiful.

All depends on how the teacher uses them. And the task of teaching mathematics is so complex that a detailed script is likely to do more harm than good.

In brief, the outcome of the formative assessment ultimately depends on the teacher's mind. The teacher attempts to understand the mathematics, the trajectories, the obstacles, the child's mind, and the principles of instruction, but in the end must use her own mind to apply all of these ideas in a personally meaningful way to the complex task of teaching the individual in the here and now.

Conclusions

The insights provided by research on the development of mathematical thinking, and the clinical interview method are essential components of formative assessment. And formative assessment based on those insights and that method is essential for successful education. Indeed, what is the alternative to obtaining detailed understanding of children's knowledge and using it to inform instruction? Or stating it another way: what are the benefits teachers can reap from ignorance concerning children's thinking and learning? It is hard to think of any. Also, it is hard to think about a sound teaching method that does not at least occasionally ask children to describe, explain and justify their thinking. At the same time, these cognitive insights and the method that facilitates their acquisition must be assimilated by the teacher's 'intermediary inventive mind,' which is the sine qua non of formative assessment.

We may think of formative assessment as a special case of teaching. Good teaching, as described for example by Lampert [2001], sometimes involves the same activities as those comprising formative assessment: understanding the mathematics, the trajectories, the child's mind, the obstacles, and using general principles of instruction to inform the teaching of a child or group of children. The intermediary inventive mind is central to teaching as well as to formative assessment.

This analysis suggests that psychology has a great deal to offer teachers, both for formative assessment and teaching in general. Psychology's major contributions to these efforts include powerful ideas about children's mathematical development and how to foster it, and the clinical interview method. We need to find ways to give away this psychology to those who most need it. One of the chief tasks for the field of mathematics education is the training, care, and sustenance of teachers' minds, as well as children's.

References

- Baroody, A.J., & Dowker, A. (Eds.) (2003). *The development of arithmetic concepts and skills: Recent research and theory*. Mahwah: Erlbaum.
- Baroody, A.J., & Ginsburg, H.P. (1983). The effects of instruction on children's understanding of the 'equals' sign. *The Elementary School Journal*, 84, 199-212.
- Baroody, A.J., Lai, M., & Mix, K.S. (2006). The development of young children's early number and operation sense and its implications for early childhood education. In B. Spodek & O. Saracho (Eds.), *Handbook of research on the education of young children*. Vol. 2 (pp. 187-221). Mahwah: Erlbaum.
- Bransford, J.D., Brown, A.L., & Cocking, R.R. (Eds.) (1998). *How people learn: Brain, mind, experience, and school*. Washington: National Academy Press.

- Clements, D.H., & McMillen, S. (1996). Rethinking 'concrete manipulatives'. *Teaching Children Mathematics*, 2, 270–279.
- Clements, D.H., & Sarama, J. (2007). Early childhood mathematics learning. In F.K. Lester (Ed.), *Second handbook of research on mathematics teaching and learning* (pp. 461–555). Charlotte: Information Age Publishing.
- Clements, D.H., Sarama, J., & DiBiase, A.M. (Eds.) (2004). *Engaging young children in mathematics: Standards for early childhood mathematics education*. Mahwah: Erlbaum.
- Darwin, C. (1955). *The expression of emotion in man and animals*. New York: Philosophical Library.
- Dewey, J. (1976). The child and the curriculum. In J.A. Boydston (Ed.), *John Dewey: The middle works, 1899–1924. Vol. 2: 1902–1903* (pp. 273–291). Carbondale: Southern Illinois University Press.
- diSessa, A.A. (2007). An interactional analysis of clinical interviewing. *Cognition and Instruction*, 25, 523–565.
- Dowker, A. (2005). *Individual differences in arithmetic: Implications for psychology, neuroscience and education*. Hove: Psychology Press.
- Fernandez, C. (2002). Learning from Japanese approaches to professional development: The case of lesson study. *Journal of Teacher Education*, 53, 393–405.
- Flanders, J. (1987). How much of the content in mathematics textbooks is new? *Arithmetic Teacher*, 35, 18–23.
- Fuchs, L.S., Fuchs, D., Karns, K., Hamlett, C.L., & Katzaroff, M. (1999). Mathematics performance assessment in the classroom: Effects on teacher planning and student problem solving. *American Educational Research Journal*, 36, 609–646.
- Geary, D.C. (1994). *Children's mathematical development: Research and practical applications*. Washington: American Psychological Association.
- Gelman, R. (1980). What young children know about numbers. *Educational Psychologist*, 15, 54–68.
- Gersten, R., Jordan, N.C., & Flojo, J.R. (2005). Early identification and interventions for students with mathematics difficulties. *Journal of Learning Disabilities*, 38, 293–304.
- Ginsburg, H.P. (1989). *Children's arithmetic: How they learn it and how you teach it* (2nd ed.). Austin: Pro Ed.
- Ginsburg, H.P. (1997). *Entering the child's mind: The clinical interview in psychological research and practice*. New York: Cambridge University Press.
- Ginsburg, H.P., & Amit, M. (2008). What is teaching mathematics to young children? A theoretical perspective and case study. *Journal of Applied Developmental Psychology*, 29, 274–285.
- Ginsburg, H.P., & Baroody, A.J. (2003). *The test of early mathematics ability* (3rd ed.). Austin: Pro Ed.
- Ginsburg, H.P., Cannon, J., Eisenband, J.G., & Pappas, S. (2006). Mathematical thinking and learning. In K. McCartney & D. Phillips (Eds.), *Handbook of early child development* (pp. 208–229). Oxford: Blackwell.
- Ginsburg, H.P., & Seo, K.H. (1999). The mathematics in children's thinking. *Mathematical Thinking and Learning*, 1, 113–129.
- Heritage, M., Kim, J., & Vendlinski, T. (2008). From evidence to action: A seamless process in formative assessment? Paper presented at the American Educational Research Association Annual Meeting, New York.
- Holt, J. (1982). *How children fail* (revised ed.). New York: Delta/Seymour Lawrence.
- James, W. (1958). *Talks to teachers on psychology; and to students on some of life's ideals*. New York: W.W. Norton & Company.
- Klein, A., & Starkey, P. (1988). Universals in the development of early arithmetic cognition. In G. Saxe & M. Gearhart (Eds.), *Children's mathematics* (pp. 5–26). San Francisco: Jossey-Bass.
- Knuth, E.J., Stephens, A.C., McNeil, N.M., & Alibali, M.W. (2006). Does understanding the equals sign matter? Evidence from solving equations. *Journal for Research in Mathematics Education*, 37, 297–312.
- Kuhn, D. (2000). Metacognitive development. *Current Directions in Psychological Science*, 9, 178–181.
- Lamon, S.J. (2007). Rational numbers and proportional reasoning: Toward a theoretical framework for research. In F.K. Lester (Ed.), *Second handbook of research on mathematics teaching and learning* (pp. 629–667). Charlotte: Information Age Publishing.
- Lampert, M. (2001). *Teaching problems and the problems of teaching*. New Haven: Yale University Press.
- Levine, S.C., Jordan, N.C., & Huttenlocher, J. (1992). Development of calculation abilities in young children. *Journal of Experimental Child Psychology*, 53, 72–103.
- Li, X., Ding, M., Capraro, M.M., & Capraro, R.M. (2008). Sources of differences in children's understandings of mathematical equality: Comparative analysis of teacher guides and student texts in China and the United States. *Cognition and Instruction*, 26, 195–217.
- Limon, M. (2001). On the cognitive conflict as an instructional strategy for conceptual change: A critical appraisal. *Learning and Instruction*, 11, 357–380.
- Ma, L. (1999). *Knowing and teaching elementary mathematics*. Mahwah: Erlbaum.

- Miller, K.F., & Parades, D.R. (1996). On the shoulders of giants: Cultural tools and mathematical development. In R.J. Sternberg & T. Ben-Zeev (Eds.), *The nature of mathematical thinking* (pp. 83–117). Mahwah: Erlbaum.
- National Council of Teachers of Mathematics (2000). *Principles and standards for school mathematics*. Reston: National Council of Teachers of Mathematics.
- National Mathematics Advisory Panel (2008). *Foundations for success: The final report of the national mathematics advisory panel*. Washington: US Department of Education.
- Nunes, T., & Bryant, P.E. (1996). *Children doing mathematics*. Oxford: Blackwell.
- Nunes, T., Schliemann, A.D., & Carraher, D.W. (1993). *Street mathematics and school mathematics*. Cambridge: Cambridge University Press.
- Piaget, J. (1952a). Autobiography. In E.G. Boring, H.S. Langfeld, H. Werner & R.M. Yerkes (Eds.), *A history of psychology in autobiography* (Vol. IV, pp. 237–256). Worcester: Clark University Press.
- Piaget, J. (1952b). *The origins of intelligence in children* (M. Cook, transl.). New York: International Universities Press.
- Piaget, J. (1976). *The child's conception of the world* (J. Tomlinson & A. Tomlinson, transl.). Totowa: Littlefield, Adams & Co.
- Piaget, J. (1985). *The equilibration of cognitive structures* (T.B. a. K.J. Thampy, transl.). Chicago: The University of Chicago Press.
- Pianta, R.C., Belsky, J., Houts, R., & Morrison, F. (2007). Opportunities to learn in America's elementary classrooms. *Science*, 315, 1795–1796.
- Post, T.R., Harel, G., Behr, M.J., & Lesh, R. (1991). Intermediate teachers' knowledge of rational number concepts. In E. Fennema, T.P. Carpenter & S.J. Lamon (Eds.), *Integrating research on teaching and learning mathematics* (pp. 177–198). Albany: SUNY Press.
- Schoenfeld, A.H. (1987). What's all the fuss about metacognition? In A.H. Schoenfeld (Ed.), *Cognitive science and mathematics education* (pp. 189–215). Hillsdale: Erlbaum.
- Schwartz, D.L., & Bransford, J.D. (1998). A time for telling. *Cognition and Instruction*, 16, 475–522.
- Seo, K.H., & Ginsburg, H.P. (2003). 'You've got to carefully read the math sentence ...': Classroom context and children's interpretations of the equals sign. In A.J. Baroody & A. Dowker (Eds.), *The development of arithmetic concepts and skills: Recent research and theory* (pp. 161–187). Mahwah: Erlbaum.
- Seo, K.H., & Ginsburg, H.P. (2004). What is developmentally appropriate in early childhood mathematics education? Lessons from new research. In D.H. Clements, J. Sarama & A.M. DiBiase (Eds.), *Engaging young children in mathematics: Standards for early childhood mathematics education* (pp. 91–104). Hillsdale: Erlbaum.
- VanLehn, K. (1990). *Mindbugs: The origins of procedural misconceptions*. Cambridge: The MIT Press.
- Vygotsky, L.S. (1978). *Mind in society: The development of higher psychological processes*. Cambridge: Harvard University Press.
- Vygotsky, L.S. (1986). *Thought and language* (A. Kozulin, transl.). Cambridge: The MIT Press.
- Zhou, Z., Peverly, S.T., & Xin, T. (2006). Knowing and teaching fractions: A cross-cultural study of American and Chinese mathematics teachers. *Contemporary Educational Psychology*, 31, 438–457.

Copyright: S. Karger AG, Basel 2009. Reproduced with the permission of S. Karger AG, Basel.
Further reproduction or distribution (electronic or otherwise) is prohibited without permission
from the copyright holder.